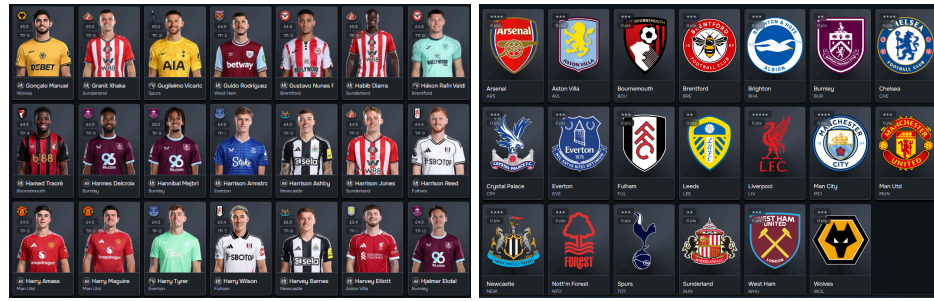


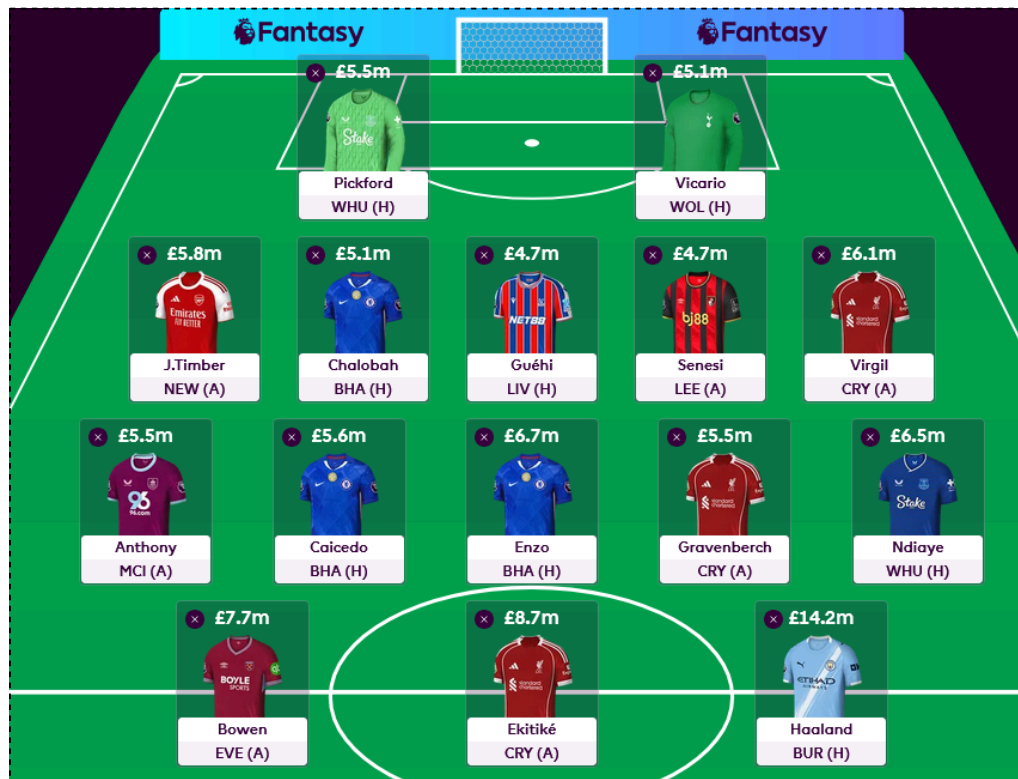
List of players from different teams



- Player list (name, club, position)
- Price
- Historical points
- Minutes/starts

Outputs & evaluation

- Recommended XI, captain, bench; formation trends
- Weekly & cumulative points; method leaderboard
- Budget sensitivity; fixed-squad sequential lineup updating
- Diagnostics/interpretation



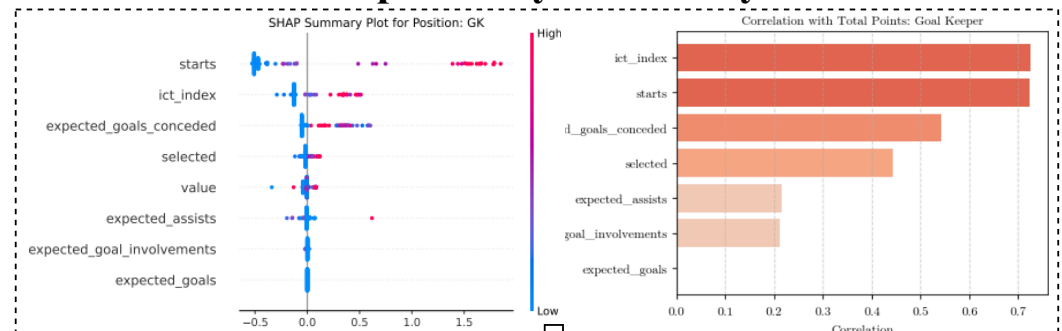
Preprocessing raw data

- Reshape/verify player–GW panel; coerce types & dedupe
- Build table schema for target GW: name, position, team, value, slot for expected points (filled by models later)
- Sort by GW per player; treat missing weeks = unavailable
- Train/Test: 1–26 → 27–38; guard against leakage (no same-GW points/bonus/BPS as inputs)

Feature selection

- For forecasting “expected total points” (simple/weighted avg, exp. smoothing (ES), ARIMA, Monte Carlo): time-series on *total_points*
- Hybrid score (Ridge per position): uses *ict_index*, *expected_goals* (*xG*), *expected_assists* (*xA*), *expected_goal_involvements* (*xGI*), *expected_goals_conceded* (*xGC*), *selected*, *starts*.
- For surrogate objectives in the XI optimizer: $xGI - xGC$, *ICT*
- *Value* is used only for budget constraints (not as a predictor).

Exploratory data analysis



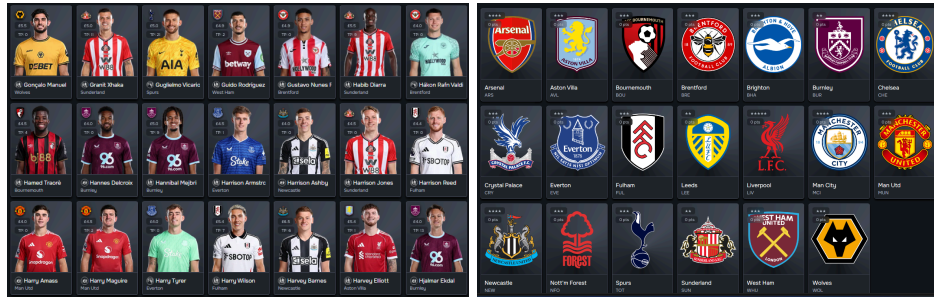
Modeling (expected points)

1. Averages/smoothing
2. ARIMA variants
3. Linear trend regression
4. Simulation: bootstrap & parametric
5. Hybrid: ridge + recent points.

Optimization (team selection)

- 0-1 Integer Programming: choose XI + captain
- Constraints: budget, formation, $\leq 3/\text{club}$, captain $\in$ XI
- Bench optimizer (remaining budget, fills to 2–5–5–3)
- Robust variant (penalize score by uncertainty)

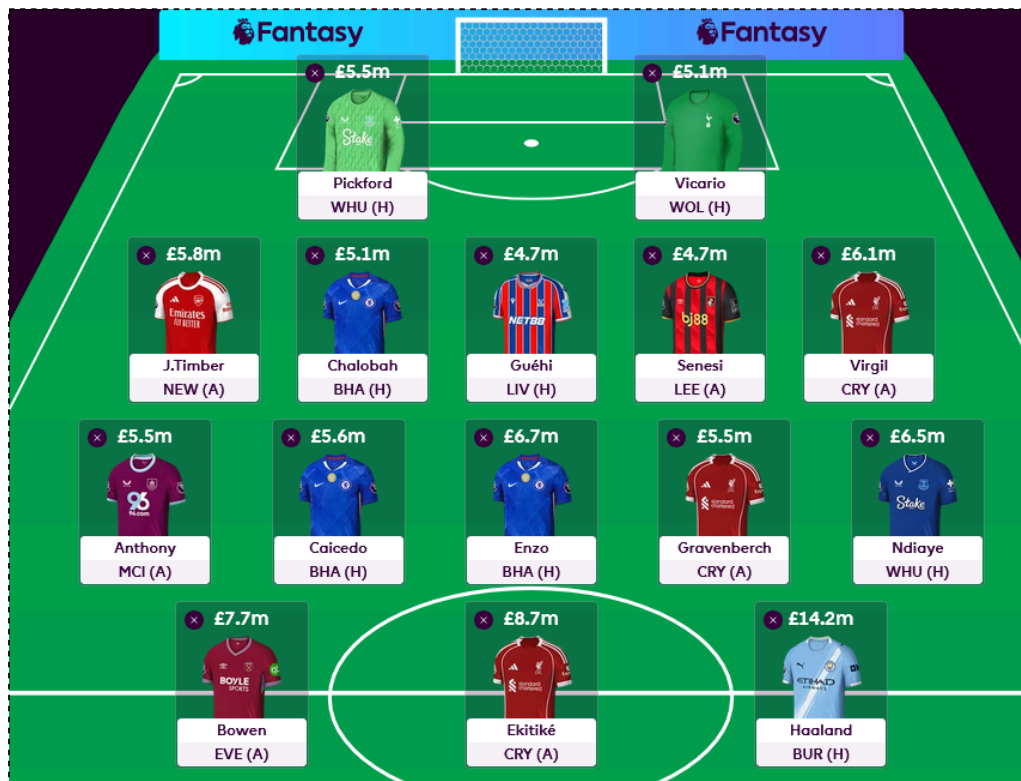
List of players from different teams



- Player list (name, club, position)
- Price
- Historical points
- Minutes/starts

Outputs & evaluation

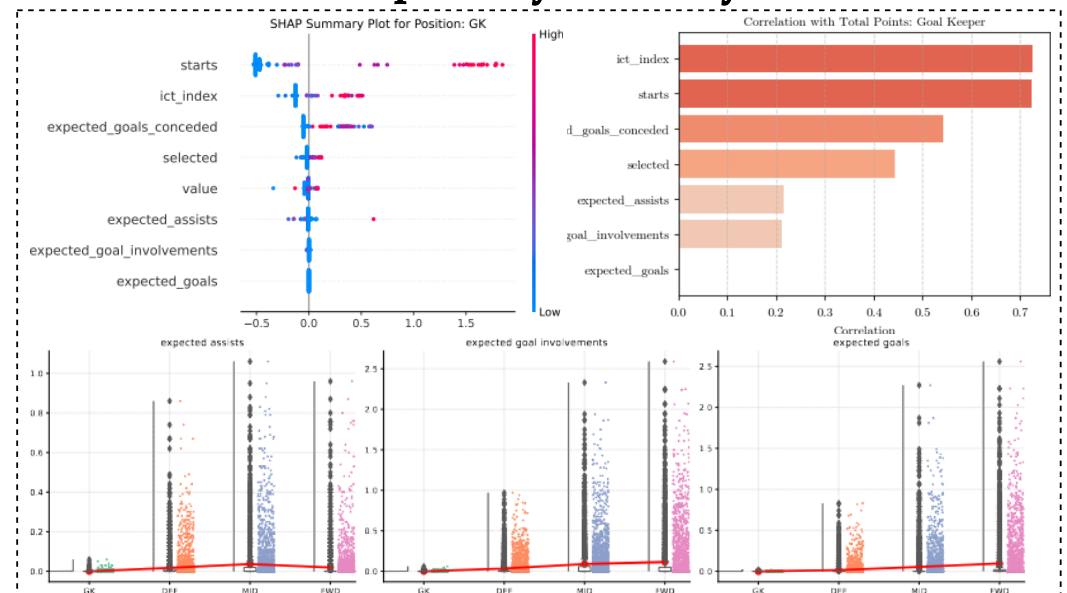
- Recommended XI, captain, bench; formation trends
- Weekly & cumulative points; method leaderboard
- Budget sensitivity; rolling-window effects
- Diagnostics/interpretation



Preprocessing and feature selection

- Reshape/verify player-GW panel; coerce types & dedupe
- Build table schema for target GW: name, position, team, value, slot for expected points (filled by models later)
- Sort by GW per player; treat missing weeks = unavailable
- Train/Test: 1-26 → 27-38; guard against leakage (no same-GW points/bonus/BPS as inputs)
- Simple/weighted avg, ARIMA, Monte Carlo: time-series on total_points
- Hybrid score: ict_index, xG, xA, xGI, xGC, selected, starts
- Surrogate objectives in the XI optimizer: xGI - xGC, ICTGI - xGC, ICT

Exploratory data analysis



Modeling (expected points)

1. Averages/smoothing
2. ARIMA variants
3. Linear trend regression
4. Simulation: bootstrap & parametric
5. Hybrid: ridge + recent points.

Optimization (team selection)

- 0-1 Integer Programming: choose XI + captain
- Constraints: budget, formation, ≤ 3 /club, captain $\in$ XI
- Bench optimizer (remaining budget, fills to 2-5-5-3)
- Robust variant (penalize score by uncertainty)